FuelSupply MCP evaluation document

Status: offline implementation verified; live market-data MVP and agent evaluation blocked.

Methodology and inputs

Versioned protocol dataset: eval/cases.jsonl (26 cases). Test-only synthetic observations are isolated in tests/protocol_server.py and tests/fuel_fixtures.py; none are verified EIA mappings. Production sessions use the actual launch entry point and empty registry. Each fixture launches a subprocess, initializes MCP, lists all five tools, reads methodology, gets the briefing prompt and invokes tools. The installed MCP SDK uses snake_case client fields; protocol serialization is SDK-owned.

Layer A/B: 74 pytest cases pass (48 FuelSupply unit/API cases; 26 inherited template/security cases). Formula, calendar, missing/nonfinite values, compatible ratios, validation, seasonal history, reconciliation, pagination, concurrency, rate limiting, permanent errors, redirects, key redaction, timeout retry exhaustion, oversized responses and cache behavior are tested. Incompatible periods are tested directly in service tests.
Layer C: real stdio protocol sessions against production and synthetic upstream fixture. Synthetic requests are in-memory MockTransport HTTP requests, never network calls.

Layer D: live EIA checks blocked, zero executed; eval/results/live.json records one skipped live-check group. EIA_API_KEY is absent and EIA hosts are not in the enforced network allowlist. The production registry has zero enabled bindings. No API route/facet/product/units combination is verified.

Layer E: attempted installed Codex CLI 0.167.0-alpha.4 using transient MCP configuration, read-only sandbox and ephemeral execution. Client initialization failed with a read-only filesystem error before any model/tool call. Logs: eval/results/agent-stderr.log and agent-events.jsonl. CLI authentication alone does not establish agent evaluation access. Agent tests executed: zero; tool-selection accuracy and model-answer correctness: unmeasured. No model configuration can be reported as executed.

Actual results

- Pytest: 74 passed, 0 failed, 0 skipped; recorded in eval/results/pytest.txt. No separate live pytest tests are implied.

- Protocol: 26 executed, 26 passed, 0 failed, 0 skipped. 15 production cases and 11 synthetic integration cases.

- Structured response validity: 25/25 non-error responses = 100%. One additional case deliberately triggers MCP input validation and is excluded from that denominator.

- Calculation checks: custom coverage 100/100 = 1 day in synthetic protocol data, insufficient seasonal history and monthly residual_present all pass; unit tests also verify a zero residual, -1 residual, leap-February conversion, invalid/zero denominator and incompatible periods. No live calculation cross-check has run.

- Error paths: unsupported/blocked coverage, aliases, input validation, fake credential 401, rate limit Retry-After, missing/withheld values, stale periods and units behave as asserted. All 26 protocol responses are checked for the fake secret string; none expose it.

- End-to-end tool-call latency (session initialization excluded), n=26: minimum 1.345 ms, median 3.349 ms, p95 reported by harness 12.050 ms, maximum 40.578 ms. These small local samples are descriptive, not a performance guarantee or official GSA target.

- Production live-requested/no-observation responses: n=14, median 1.467 ms. No upstream data were fetched. The validation error is excluded from mode groups.

- Synthetic uncached/live-requested responses (including error paths): n=9, median 5.535 ms. Synthetic cached responses: n=2, median 4.898 ms. Production cached measurements: n=0. These are not live EIA cold/cache benchmarks.

- Upstream request count: real EIA 0; synthetic HTTP transport 72. Request counts do not count internal MCP calls as EIA requests.

Environment and reproducibility

Python 3.12.14; dependencies {"fastmcp": "4.0.11", "httpx": "0.28.1", "mcp": "2.3.0", "pydantic": "2.13.5"}. Registry 0.1.0. Base/template git revision f39249f1768309e3ef8b31742390a4f3a656b5f3; runtime source SHA256 d802dc71c0c3400e60b082b18c2f2275c6c54594c30b0154d5af97b8c391abb7. Working tree contains the implementation; base revision is not falsely described as an implementation commit. uv.lock captures the full dependency set. Raw per-case tool outputs, latency and initialization timings are in eval/results/protocol.json. CI is defined but not run remotely; Python3.11 and container reproduction are unverified.

From /workspace/fuelsupply-mcp:

```bash
export UV_CACHE_DIR=/tmp/fuelsupply-uv-cache
export UV_PYTHON_INSTALL_DIR=/tmp/fuelsupply-python
uv sync --locked --group dev --python "$(command -v python)"
.venv/bin/ruff check .
.venv/bin/ruff format --check .
.venv/bin/pytest tests -q
.venv/bin/python eval/run_protocol_eval.py
.venv/bin/python eval/run_live_checks.py
.venv/bin/python scripts/build_submission.py
```

Live-check exit 2 means blocked, not success. Protocol expectations intentionally target the empty production registry; version the dataset before enabling live coverage. The server launch is /workspace/fuelsupply-mcp/.venv/bin/fuelsupply-mcp and uses stdio independently of Codex.

Security considerations

Fixed HTTPS EIA origin, relative-route validation and disabled redirects restrict outbound access. Server-side credentials are never tool inputs. Error bodies and request URLs are suppressed; numeric observations are sanitized. Timeouts, bounded concurrency/pages/retries, response-size cap and an expiring LRU limit resource use. Cache preserves retrieval timestamps and mode. No silent replay fallback, shell/file tool, purchasing operation or external tracing is enabled. Managed proxy trust is inherited. See docs/security.md for residual threats; no production, agency or compliance certification is claimed.

Failures, lessons and recommendations

Initial protocol evaluation failed because the current SDK exposes snake_case client attributes; the harness was repaired against inspected SDK signatures. A fixture regenerated its verification timestamp between requests and accidentally bypassed the cache; it now reuses the same binding. Synthetic cutoff scenarios initially filtered their own test observations; the fixture was corrected to use periods preceding the cutoff. Initial failures remain in eval/results/initial-protocol-failures.json; final results supersede them.

The optional template Phoenix/LangChain guide was inspected; it is not an official mandatory harness. It would introduce model/tracing dependencies and was not enabled. Deterministic protocol testing does not replace agent usability testing.

Next: configure credentials and permitted EIA hosts, inspect a small metadata subset, validate each mapping and accounting component against authenticated responses/published tables, then run live protocol/calculation demos and an actual MCP-capable model client. Replace this deck with the official template after downloading it in a permitted environment. Do not deploy or submit without explicit authorization.

Deliverable validation

A real local PPTX and DOCX are generated and reopened using their libraries. Rendering/visual QA is recorded separately in submission/artifact-validation.json; document reopen alone is not slide visual QA. Official PPTX/DOCX binary templates could not be downloaded here (proxy 403); local files do not claim template compliance. Requirements matrix and submission checklist list remaining gaps.